Machine Learning Models that Enable Interactions between Mind and World

2025-2026 Dissertation Progress Report

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1 Introduction

Multimodal machine learning has made significant progress in recent years [6, 7, 8, 9, 10]. Modern models are increasingly capable of processing and integrating heterogeneous data sources—such as text, images, audio, and video—enabling a wide range of applications in generation, reasoning, and interaction.

When translating between different modalities, language is usually treated as the primary communication modality. For example, many generative models adopt a Text-to-X paradigm [11, 12, 13] (e.g., text-to-image generation), where language serves as the interface through which users specify intent. Conversely, X-to-Text paradigms are also common [7, 10, 14]. Vision large language models, for instance, take images together with linguistic instructions as input and produce responses in natural language, supporting tasks such as visual question answering and visual reasoning. In everyday life, voice assistants (e.g., Siri and Alexa) further exemplify this design choice: users communicate intentions through language, and systems execute actions accordingly. However, this prevailing design raises a fundamental question: *is language truly the most efficient and effective communication modality?*

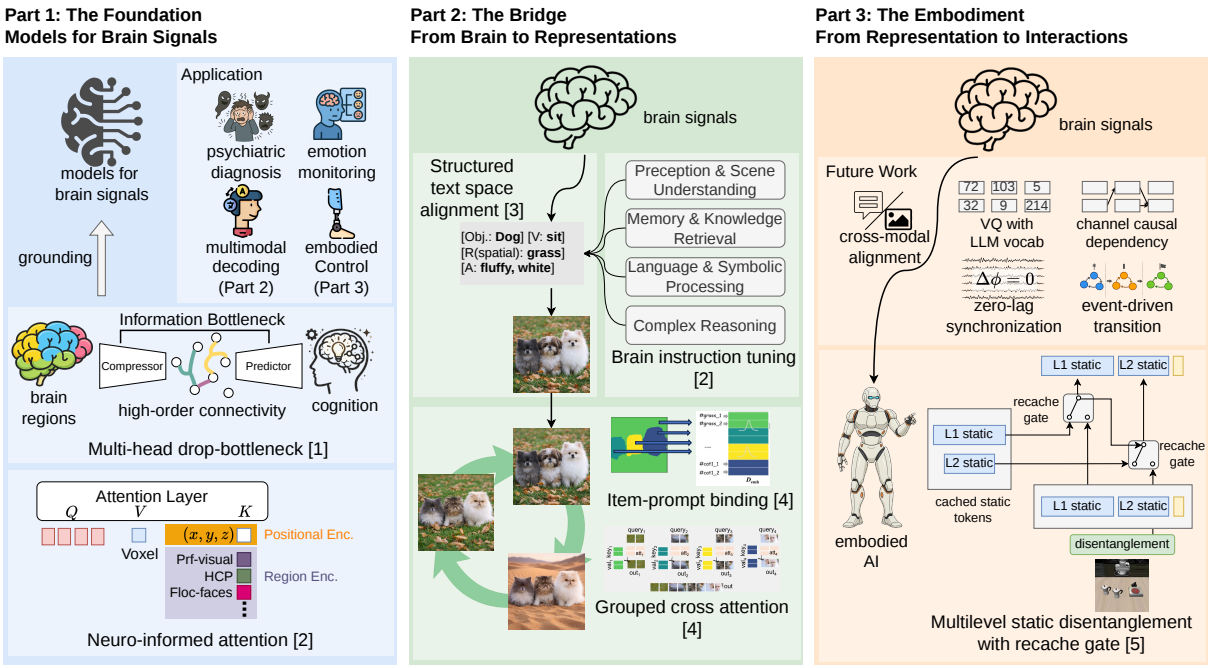


Figure 1: The overview of the research plan. [1, 2, 3, 4, 5]

There are several limitations to using language as the primary communication modality. First, verbal and

written language are inherently low-bandwidth. Empirical studies suggest that speaking produces approximately 100–150 words per minute, while human thought unfolds at an estimated rate of 400–800 words per minute [15], indicating a substantial mismatch between internal cognition and external expression. This bottleneck becomes evident in practical systems such as text-to-image generation models, where users often must craft lengthy and detailed prompts to specify visual attributes precisely—a process that can be both time-consuming and cognitively inefficient.

Second, not all forms of experience or meaning are readily expressible in language. Consider Composition VII, painted by Wassily Kandinsky in 1913: its abstract structure and affective intensity resist precise linguistic description. This limitation is not merely a matter of insufficient eloquence, but rather reflects the existence of phenomena that cannot be fully articulated in language and can only be shown. As Ludwig Wittgenstein famously stated, "Whereof one cannot speak, thereof one must be silent." [16]

Motivated by these observations, we propose to build machine learning models that enable more direct communication between Mind and World. Achieving this paradigm requires addressing three fundamental challenges:

- Developing machine learning models that incorporate neural priors to better reflect brain organization, thereby improving interpretability, data efficiency, and performance on mind-related tasks.
- Learning representations that align brain activity with high-level semantic and perceptual spaces under limited supervision.
- Bridging the gap between neural representations and continuous action spaces, and enabling real-time, closed-loop interaction with external systems.

To address these challenges, the dissertation is structured into three parts. First, I develop neuroscience-informed models for brain signals. Second, I design representation learning frameworks that bridge neural activity with semantic and visual spaces. Third, I build integrative systems that connect neural signals to embodied agents, enabling direct neural-to-action interaction.

2 Research Objectives

This research plan progresses in three stages, moving from neural modeling to semantic alignment and finally to embodied interaction:

- (Part 1) The Foundation: Models for Brain Signals
- (Part 2) The Bridge: From Signals to Representations
- (Part 3) The Embodiment: From Representations to Interaction

Each stage builds upon the previous one, forming a coherent path from neural understanding to actionable interactions.

(Part 1) The Foundation: Models for Brain Signals

Existing works usually simply migrate methods from CV and NLP [17, 18, 18, 19], without much architectural designs or training objectives that incorporate domain inductives bias. Such a paradigm, on the one hand, cannot offer neuroscientific insights that could benefit neuroscience research; on the other hand, lead to suboptimal performance. This part focuses on the "mind" side of the problem, developing machine learning models that are informed by brain structure and neural mechanisms. The goal is twofold: (1) to improve our understanding of how neural activity encodes cognitive and perceptual information, and (2) to leverage these insights to achieve stronger predictive performance on downstream tasks.

Prior works HyBRiD Most existing methods rely primarily on pairwise relationships between brain regions, overlooking higher-order interactions that are essential for complex cognitive functions. Unlike pairwise relationships, high-order ones reside in an exponential space, making them challenging to identify. To address

this limitation, my prior work introduced HyBRiD [1], a framework for identifying high-order relationships among brain regions from fMRI data, and benefiting predictive tasks. We propose an information-theoretic objective that seeks relationships that are maximally informative yet minimally redundant for end-to-end training. By explicitly modeling these high-order structures, HyBRiD not only provides more interpretable representations of brain connectivity but also achieves superior performance on downstream tasks such as intelligence quotient prediction. This line of work establishes a foundation for neural representations that better reflect the true complexity of brain organization and cognition.

MindLLM Existing methods [20, 21] on fMRI multimodal decoding mostly rely on a preprocessing step: selecting responsive voxels by comparing the task-based fMRI to the resting-state fMRI. The selection typically results in higher performance compared to using whole-brain data. However, the varying number and irregular spatial distribution of selected voxels across subjects pose significant challenges for developing a unified architecture. To this end, recent studies [22, 23] have proposed pooling or sampling voxels to standardize input dimensions. However, these methods still suffer from the loss of spatial information and uneven representations of certain areas, ultimately compromising performance. In MindLLM [2], we propose a neuroscience-informed attention, where we use learnable queries to enable dynamic feature extraction by leveraging both spatial information and neuroscientific priors of voxels. By jointly leveraging structural information and domain knowledge, our framework achieves strong cross-subject decoding performance and demonstrates robust generalization to unseen subjects.

(Part 2) The Bridge: From Signals to Representations

In Part 1, we focused on modeling brain signals within the neural modality itself. In this part, we move beyond intra-modal modeling and investigate how to learn representations of brain activity that align with world-level semantics. While multimodal alignment has achieved remarkable progress in domains such as vision and language, brain signals remain substantially underexplored. Compared to image–text datasets, brain datasets are scarce, expensive to acquire, and highly variable across subjects. Therefore, building a reliable bridge from brain activity to semantic representations requires not only architectural designs grounded in cognitive principles but also carefully constructed datasets and proposed training paradigms that can operate under limited supervision and cross-subject variability.

Prior works MindLLM Existing brain-to-text decoding approaches primarily focus on stimulus reconstruction. While these methods have demonstrated impressive perceptual recovery, they suffer from limited task diversity, poor generalization across subjects, and weak alignment with higher-level cognition [21, 22, 20]. MindLLM [2] overcomes these limitations by directly mapping fMRI signals into a large language model using neuroscience-informed attention and Brain Instruction Tuning, enabling subject-agnostic and task-flexible decoding. By supporting reasoning, memory retrieval, and novel task adaptation beyond perceptual reconstruction. MindLLM establishes the first step of the bridge: mapping neural signals into a high-level semantic space, which provides a unified representational interface for further applications and tasks.

PRISM While language provides a robust semantic anchor, full perceptual reconstruction requires a tighter coupling between semantics and fine-grained visual information. Prior fMRI-to-image reconstruction methods typically rely on vision-centric or joint vision–language latent spaces [24, 21, 22], which are poorly aligned with the structured and compositional nature of neural representations, leading to limited reconstruction fidelity and object-level errors. PRISM [3] addresses this limitation by projecting fMRI signals into a structured text space that better matches neural organization, and by explicitly modeling objects, attributes, and relationships through object-centric diffusion and attribute–relationship search. This design enables more accurate and semantically coherent neural-driven visual reconstruction, and moves the bridge from abstract semantics to coherent, high-fidelity visual reconstruction.

D-Edit Even with structured alignment, decoded signals are rarely perfect; a "post-edit" phase is essential to bridge the gap between neural reconstructions and intended imagery. Existing diffusion-based image editing methods [25, 26, 27] suffer from entangled prompt semantics, where editing individual words often causes unintended global changes, and from unreliable mask-based control that diffusion models tend to ignore, leading to inharmonic results. To address these limitations, we propose D-Edit [4], a framework

that disentangles item-level control in diffusion models by decomposing image–prompt interactions into multiple item–prompt pairs. Each item is associated with a learned prompt, enabling precise and flexible manipulation through prompt editing. This design supports diverse editing operations—including image-, text-, and mask-based editing as well as item removal—within a unified framework, and potentially enables flexible neural-driven editing.

(Part 3) The Emodiment: From Representations to Interactions

While Part 1 established the foundation for neural modeling and Part 2 bridged those signals to semantic and visual representations, Part 3 addresses the final challenge: embodiment. To move from passive decoding to active communication, we must enable direct, high-bandwidth interactions between the brain and the physical world. Current Embodied AI models face challenges of both applicability and integration. On one hand, existing Vision-Language-Action (VLA) models are computationally too heavy to handle the high-throughput, real-time demands of neural-driven control. On the other hand, there remains a fundamental architectural gap between the spaces of embodied agents and the brain signals. This part of the research proposes solutions for both the applicability of embodied agents and the alignment of neural signals with continuous action spaces.

Prior works SD-VLA Existing Vision-Language-Action models (VLAs) [28, 29, 30] face two major challenges: limited long-horizon context and inefficient inference due to the quadratic attention complexity and large parameter counts. Our work is motivated by the observation that much of the visual information in a trajectory remains static across timesteps (e.g., the background). Leveraging this property, we propose SD-VLA [5], a framework that disentangles visual inputs into multi-level static and dynamic tokens, which enables (1) retaining a single copy of static tokens across frames to significantly reduce context length, and (2) reusing the key-value (KV) cache of static tokens through a lightweight recache gate that updates only when necessary. This design enables temporal-dependency modeling and efficient inference, thereby improving its practical usability.

On the brain side, we develop an EEG foundation model with a world-model pretraining objective, where the model predicts neural representations of imagined future states from current EEG activity. This formulation aligns with motor imagery and captures motor-relevant temporal dynamics rather than spurious artifacts such as volume conduction. Initial experiments show promising results on downstream motor control benchmarks.

Future Works Works Building on the EEG foundation model, the current world-model pretraining is effective but its downstream performance still has room for improvement. We plan to incorporate latest architectural advances to further strengthen the learned representations. Besides, we plan to collect a novel EEG-Robotic Arm dataset to demonstrate the model's ability to decode neural signals into continuous action, completing the transition from internal thought to physical embodiment.

3 Future Plan

Year 5 (including summer)

In Year 5, I will focus on dataset collection, model adaptation for continuous control, and dissertation completion.

- **(Jun-Jul) EEG dataset collection for continuous brain-controlled robotics.** Most existing EEG datasets emphasize classification tasks, while regression settings typically focus on predicting human limb kinematics. There is limited work on continuous control of external devices (e.g., robotic arms or neuroprosthetics) directly from brain signals. To address this gap, I will collect a new EEG dataset involving continuous brain-driven control of robotic manipulators. Using Robosuite as the simulation platform, participants will perform tasks such as object manipulation, spatial arrangement, and binary control operations (e.g., turning switches on/off). The dataset will emphasize continuous trajectories and fine-grained motor intention signals to support closed-loop brain–machine interaction.

- **(Aug) Fine-tuning and reinforcement learning integration.** I will fine-tune the pretrained motor-control foundation model on the newly collected dataset. In addition, I will explore integration with state-of-the-art reinforcement learning algorithms for continuous control. Both the actor and reward models may be conditioned on neural signals, enabling brain-informed policy learning and reward shaping. This framework aims to bridge representation learning and adaptive control for brain–robot interaction.
- **(Sep-Oct) Dissertation writing and defense preparation.** I will complete the writing of the dissertation, synthesize experimental findings across all project phases, and prepare presentation materials for the doctoral defense.
- **(Sep) Dissertation defense and revision.** I will defend the dissertation and incorporate revisions based on committee feedback to finalize the document for submission.

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